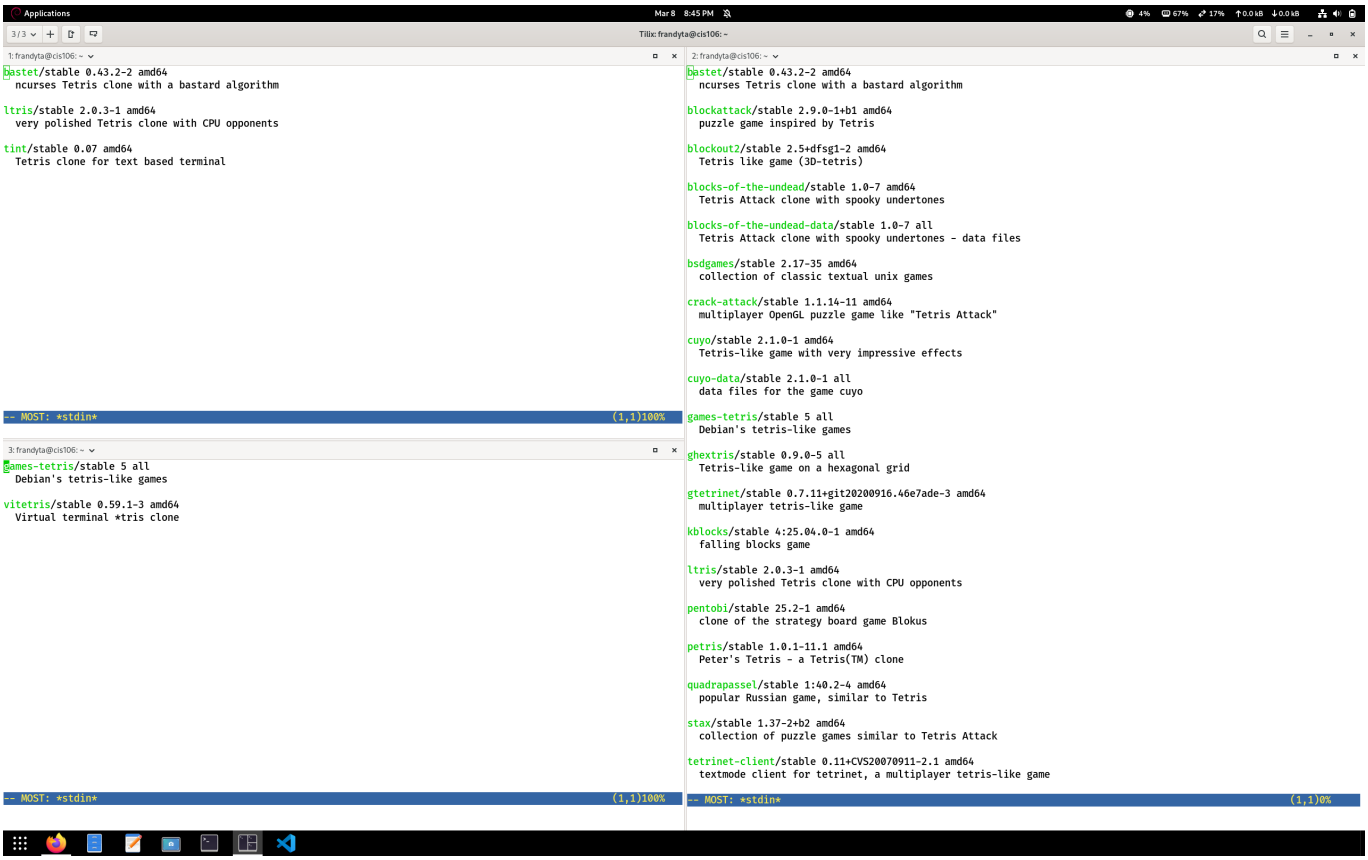
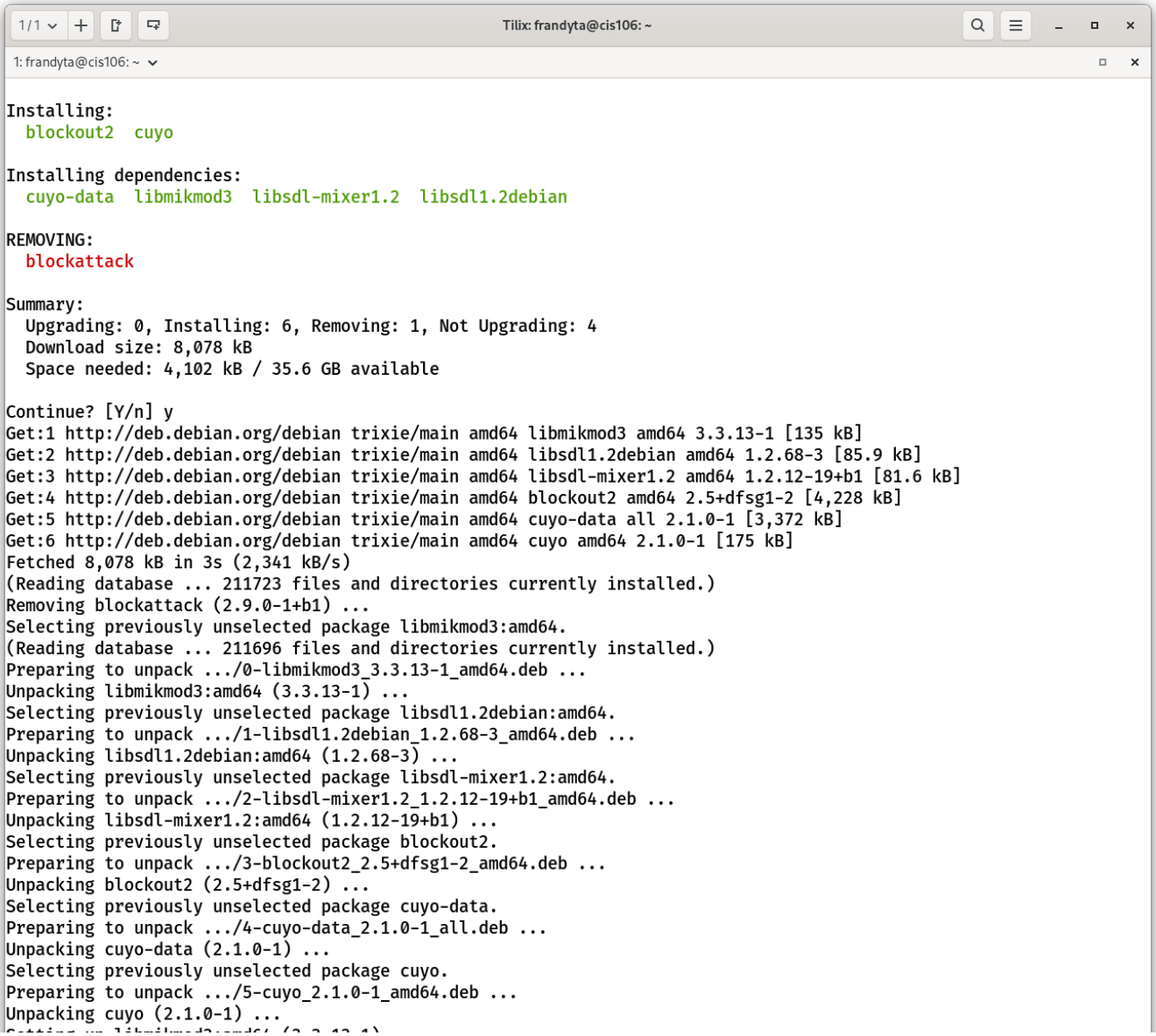
Lab 4 Submission

Question 1


Applications
Mar 8 8:45 PM
Tilix: frandyta@cis106: ~
1: frandyta@cis106: ~
bastet/stable 0.43.2-2 amd64
ncurses Tetris clone with a bastard algorithm
ltris/stable 2.0.3-1 amd64
very polished Tetris clone with CPU opponents
tint/stable 0.07 amd64
Tetris clone for text based terminal
-- MOST: *stdin* (1,1)100%
2: frandyta@cis106: ~
bastet/stable 0.43.2-2 amd64
ncurses Tetris clone with a bastard algorithm
blockattack/stable 2.9.0-1+b1 amd64
puzzle game inspired by Tetris
blockout2/stable 2.5+dfsg1-2 amd64
Tetris like game (3D-tetris)
blocks-of-the-undead/stable 1.0-7 amd64
Tetris Attack clone with spooky undertones
blocks-of-the-undead-data/stable 1.0-7 all
Tetris Attack clone with spooky undertones - data files
bsdgames/stable 2.17-35 amd64
collection of classic textual unix games
crack-attack/stable 1.1.14-11 amd64
multiplayer OpenGL puzzle game like "Tetris Attack"
cuyo/stable 2.1.0-1 amd64
Tetris-like game with very impressive effects
cuyo-data/stable 2.1.0-1 all
data files for the game cuyo
games-tetris/stable 5 all
Debian's tetris-like games
ghextris/stable 0.9.0-5 all
Tetris-like game on a hexagonal grid
gtetrinet/stable 0.7.11+git20200916.46e7ade-3 amd64
multiplayer tetris-like game
kblocks/stable 4:25.04.0-1 amd64
falling blocks game
ltris/stable 2.0.3-1 amd64
very polished Tetris clone with CPU opponents
pentobi/stable 25.2-1 amd64
clone of the strategy board game Blokus
petris/stable 1.0.1-11.1 amd64
Peter's Tetris - a Tetris(TM) clone
quadrapassel/stable 1:40.2-4 amd64
popular Russian game, similar to Tetris
stax/stable 1.37-2+b2 amd64
collection of puzzle games similar to Tetris Attack
tetrinet-client/stable 0.11+CVS20070911-2.1 amd64
textmode client for tetrinet, a multiplayer tetris-like game
-- MOST: *stdin* (1,1)0%
3: frandyta@cis106: ~
games-tetris/stable 5 all
Debian's tetris-like games
vitetris/stable 0.59.1-3 amd64
Virtual terminal *tris clone
-- MOST: *stdin* (1,1)100%


1/1
Tilix: frandyta@cis106: ~
1: frandyta@cis106: ~
Installing:
blockout2 cuyo
Installing dependencies:
cuyo-data libmikmod3 libsdl-mixer1.2 libsdl1.2debian
REMOVING:
blockattack
Summary:
Upgrading: 0, Installing: 6, Removing: 1, Not Upgrading: 4
Download size: 8,078 kB
Space needed: 4,102 kB / 35.6 GB available
Continue? [Y/n] y
Get:1 http://deb.debian.org/debian trixie/main amd64 libmikmod3 amd64 3.3.13-1 [135 kB]
Get:2 http://deb.debian.org/debian trixie/main amd64 libsdl1.2debian amd64 1.2.68-3 [85.9 kB]
Get:3 http://deb.debian.org/debian trixie/main amd64 libsdl-mixer1.2 amd64 1.2.12-19+b1 [81.6 kB]
Get:4 http://deb.debian.org/debian trixie/main amd64 blockout2 amd64 2.5+dfsg1-2 [4,228 kB]
Get:5 http://deb.debian.org/debian trixie/main amd64 cuyo-data all 2.1.0-1 [3,372 kB]
Get:6 http://deb.debian.org/debian trixie/main amd64 cuyo amd64 2.1.0-1 [175 kB]
Fetched 8,078 kB in 3s (2,341 kB/s)
(Reading database ... 211723 files and directories currently installed.)
Removing blockattack (2.9.0-1+b1) ...
Selecting previously unselected package libmikmod3:amd64.
(Reading database ... 211696 files and directories currently installed.)
Preparing to unpack .../0-libmikmod3_3.3.13-1_amd64.deb ...
Unpacking libmikmod3:amd64 (3.3.13-1) ...
Selecting previously unselected package libsdl1.2debian:amd64.
Preparing to unpack .../1-libsdl1.2debian_1.2.68-3_amd64.deb ...
Unpacking libsdl1.2debian:amd64 (1.2.68-3) ...
Selecting previously unselected package libsdl-mixer1.2:amd64.
Preparing to unpack .../2-libsdl-mixer1.2_1.2.12-19+b1_amd64.deb ...
Unpacking libsdl-mixer1.2:amd64 (1.2.12-19+b1) ...
Selecting previously unselected package blockout2.
Preparing to unpack .../3-blockout2_2.5+dfsg1-2_amd64.deb ...
Unpacking blockout2 (2.5+dfsg1-2) ...
Selecting previously unselected package cuyo-data.
Preparing to unpack .../4-cuyo-data_2.1.0-1_all.deb ...
Unpacking cuyo-data (2.1.0-1) ...
Selecting previously unselected package cuyo.
Preparing to unpack .../5-cuyo_2.1.0-1_amd64.deb ...
Unpacking cuyo (2.1.0-1) ...

```
Setting up libmikmod3:amd64 (3.3.13-1) ...
Setting up libsdl1.2debian:amd64 (1.2.68-3) ...
Setting up libsdl-mixer1.2:amd64 (1.2.12-19+b1) ...
Setting up cuyo-data (2.1.0-1) ...
Setting up cuyo (2.1.0-1) ...
Setting up blockout2 (2.5+dfsg1-2) ...
Processing triggers for hicolor-icon-theme (0.18-2) ...
Processing triggers for gnome-menus (3.36.0-3) ...
Processing triggers for libc-bin (2.41-12+deb13u1) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for mailcap (3.74) ...
Processing triggers for desktop-file-utils (0.28-1) ...
frandyta@cis106:~$
```

Tilix: frandyta@cis106: ~

1: frandyta@cis106: ~

```
frandyta@cis106:~$ sudo apt remove cuyo
The following packages were automatically installed and are no longer required:
  cuyo-data  libopusfile0  libsdl2-image-2.0-0  libsdl2-ttf-2.0-0
  libfmt10   libphysfs1    libsdl2-mixer-2.0-0  libxmp4
Use 'sudo apt autoremove' to remove them.

REMOVING:
  cuyo

Summary:
  Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 4
  Freed space: 489 kB

Continue? [Y/n] n
Abort.
frandyta@cis106:~$ sudo apt remove cuyo
The following packages were automatically installed and are no longer required:
  cuyo-data  libopusfile0  libsdl2-image-2.0-0  libsdl2-ttf-2.0-0
  libfmt10   libphysfs1    libsdl2-mixer-2.0-0  libxmp4
Use 'sudo apt autoremove' to remove them.

REMOVING:
  cuyo

Summary:
  Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 4
  Freed space: 489 kB

Continue? [Y/n] y
(Reading database ... 212793 files and directories currently installed.)
Removing cuyo (2.1.0-1) ...
Processing triggers for hicolor-icon-theme (0.18-2) ...
Processing triggers for gnome-menus (3.36.0-3) ...
Processing triggers for mailcap (3.74) ...
Processing triggers for desktop-file-utils (0.28-1) ...
frandyta@cis106:~$ sudo apt purge cuyo
The following packages were automatically installed and are no longer required:
  cuyo-data  libopusfile0  libsdl2-image-2.0-0  libsdl2-ttf-2.0-0
  libfmt10   libphysfs1    libsdl2-mixer-2.0-0  libxmp4
Use 'sudo apt autoremove' to remove them.

REMOVING:
  cuyo*

Summary:
  Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 4
  Space needed: 0 B / 35.6 GB available

Continue? [Y/n] y

(Reading database ... 212785 files and directories currently installed.)
Purging configuration files for cuyo (2.1.0-1) ...
frandyta@cis106:~$
```

```
frandyta@cis106:~$ sudo apt clean && sudo apt autoclean && sudo apt autoremove && sudo apt autopurge
REMOVING:
  cuyo-data  libopusfile0  libsdl2-image-2.0-0  libsdl2-ttf-2.0-0
  libfmt10   libphysfs1    libsdl2-mixer-2.0-0  libxmp4

Summary:
  Upgrading: 0, Installing: 0, Removing: 8, Not Upgrading: 4
  Freed space: 7,495 kB

Continue? [Y/n] y
(Reading database ... 212785 files and directories currently installed.)
Removing cuyo-data (2.1.0-1) ...
Removing libfmt10:amd64 (10.1.1+ds1-4) ...
Removing libsdl2-mixer-2.0-0:amd64 (2.8.1+dfsg-2) ...
Removing libopusfile0:amd64 (0.12-4+b3) ...
Removing libphysfs1:amd64 (3.0.2-6+b2) ...
Removing libsdl2-image-2.0-0:amd64 (2.8.8+dfsg-1) ...
Removing libsdl2-ttf-2.0-0:amd64 (2.24.0+dfsg-2) ...
Removing libxmp4:amd64 (4.6.2-2) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for libc-bin (2.41-12+deb13u1) ...
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 4
frandyta@cis106:~$
```

Question 2

Open lab4.script.sh ~/Script

```
#!/bin/bash

```

1/1 Tilix: frandyta@cis106: ~/Script

1: frandyta@cis106: ~/Script

```
frandyta@cis106:~$ cd Scrpt
bash: cd: Scrpt: No such file or directory
frandyta@cis106:~$ cd Script
frandyta@cis106:~/Script$ bash lab4.script.sh
frandyta@cis106:~/Script$
```

Open lab4-script2.sh ~/Script

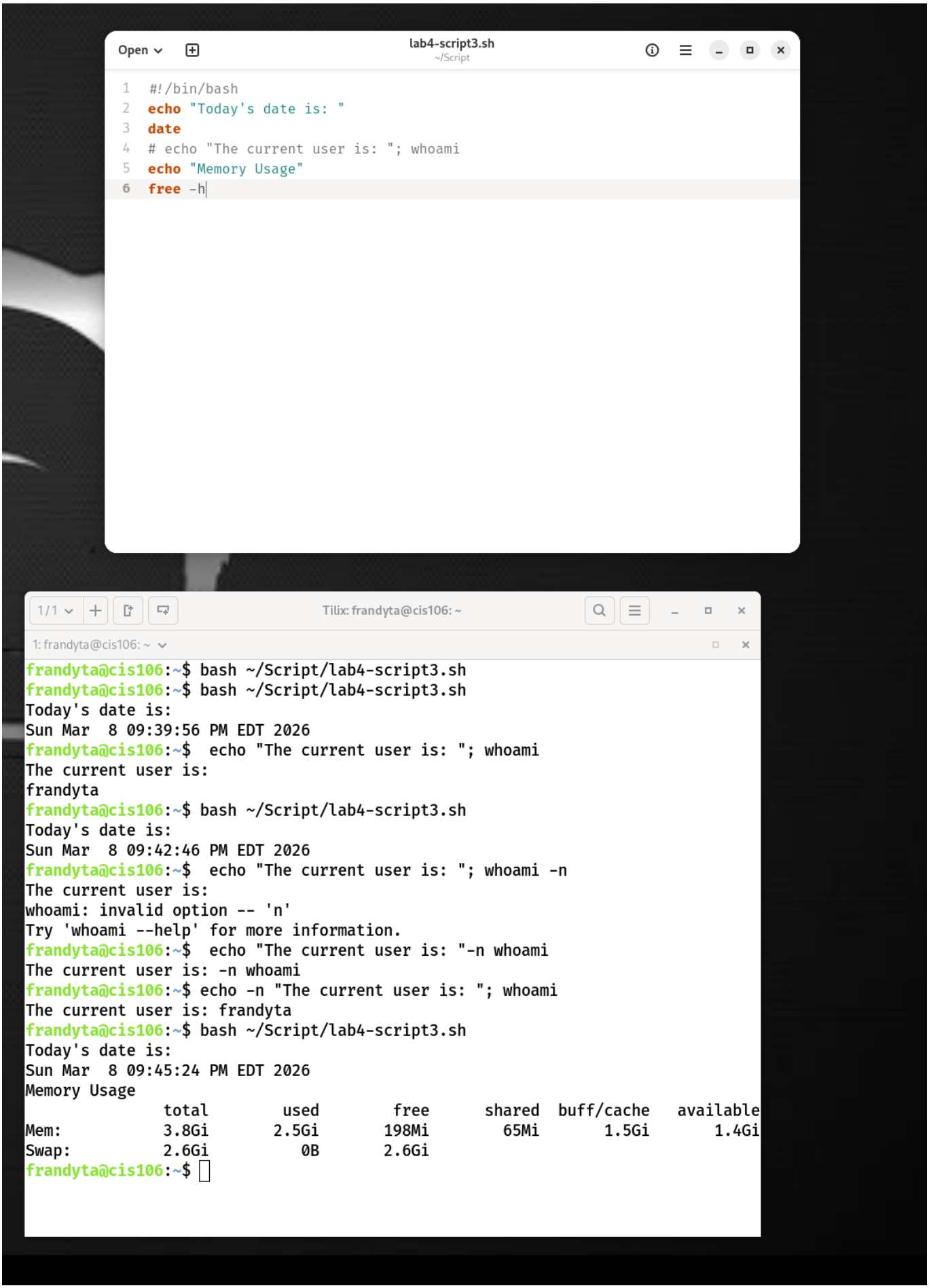
lab4.script.sh lab4-script2.sh

```bash
#!/bin/bash
echo "Hello World"
echo -e "Name:\tFrandy"
echo -e "Age:\t21"
echo -e "Colo:r\tBlue\nGame:\tTetris"
```

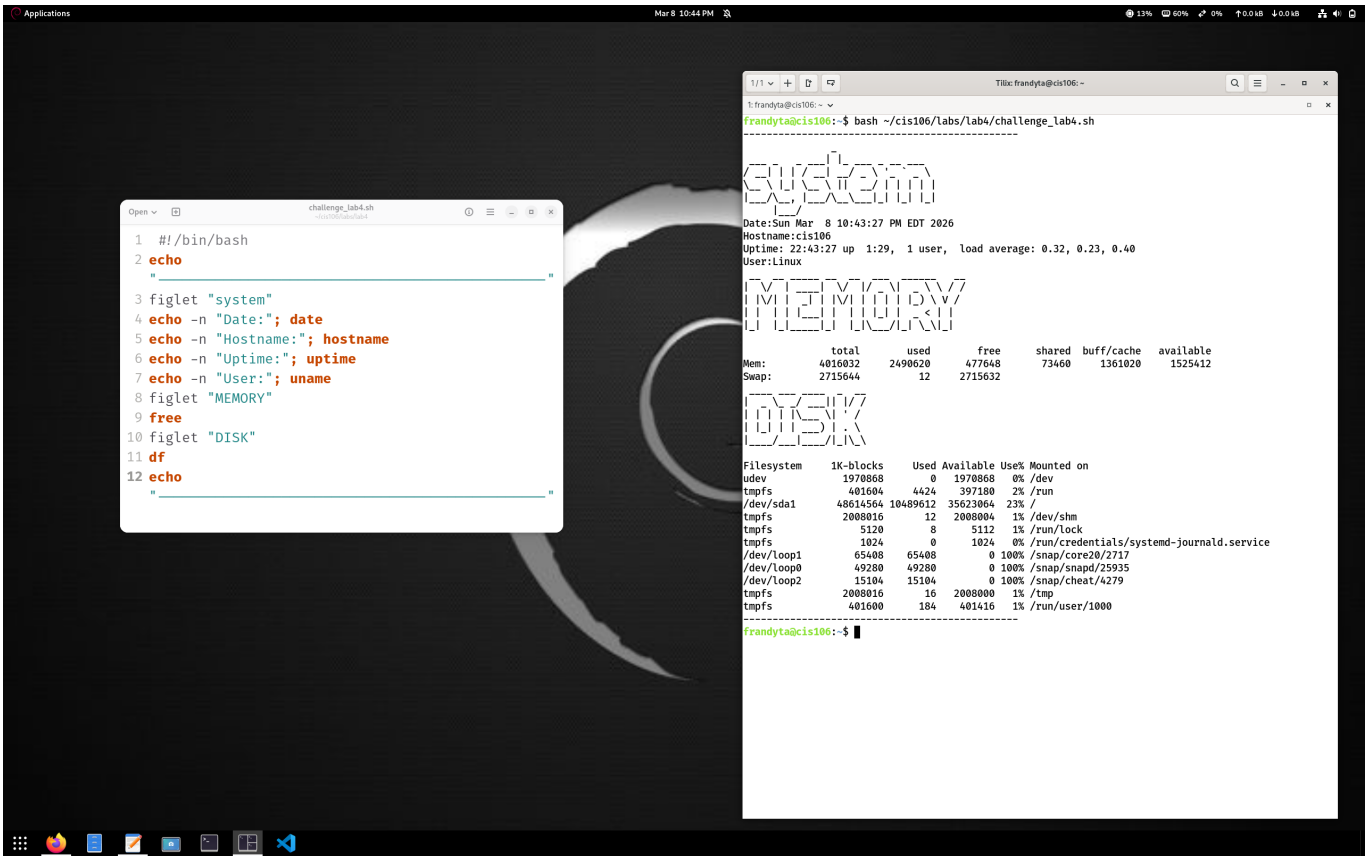
Tilix: frandyta@cis106: ~

1: frandyta@cis106: ~

```
frandyta@cis106:~$ bash ~/Script/lab4-script2.sh
Hello World
frandyta@cis106:~$ bash ~/Script/lab4-script2.sh
Hello World
Name:	Frandy
Age:	21
frandyta@cis106:~$ bash ~/Script/lab4-script2.sh
Hello World
Name:	Frandy
Age:	21
Colo:r	Blue
Game:	Tetris
frandyta@cis106:~$
```


Open
lab4-script3.sh
~/Script
#!/bin/bash
echo "Today's date is: "
date
echo "The current user is: "; whoami
echo "Memory Usage"
free -h
1/1
Tilix: frandyta@cis106: ~
1: frandyta@cis106: ~
frandyta@cis106:~$ bash ~/Script/lab4-script3.sh
frandyta@cis106:~$ bash ~/Script/lab4-script3.sh
Today's date is:
Sun Mar 8 09:39:56 PM EDT 2026
frandyta@cis106:~$ echo "The current user is: "; whoami
The current user is:
frandyta
frandyta@cis106:~$ bash ~/Script/lab4-script3.sh
Today's date is:
Sun Mar 8 09:42:46 PM EDT 2026
frandyta@cis106:~$ echo "The current user is: "; whoami -n
The current user is:
whoami: invalid option -- 'n'
Try 'whoami --help' for more information.
frandyta@cis106:~$ echo "The current user is: "-n whoami
The current user is: -n whoami
frandyta@cis106:~$ echo -n "The current user is: "; whoami
The current user is: frandyta
frandyta@cis106:~$ bash ~/Script/lab4-script3.sh
Today's date is:
Sun Mar 8 09:45:24 PM EDT 2026
Memory Usage
              total   used   free   shared buff/cache   available
Mem: 3.8Gi   2.5Gi   198Mi   65Mi   1.5Gi   1.4Gi
Swap: 2.6Gi   0B   2.6Gi
frandyta@cis106:~$

Question 3


Applications
Mar 8 10:44 PM
Tilix: frandyta@cis106: ~
1: frandyta@cis106: ~
frandyta@cis106:~$ bash ~/cis106/labs/lab4/challenge_lab4.sh
Date:Sun Mar 8 10:43:27 PM EDT 2026
Hostname:cis106
Uptime: 22:43:27 up 1:29, 1 user, load average: 0.32, 0.23, 0.40
User:Linux
Mem: 4016032 2490620 477648 73460 1361020 1525412
Swap: 2715644 12 2715632
Filesystem 1K-blocks Used Available Use% Mounted on
udev 1970868 0 1970868 0% /dev
tmpfs 401604 4424 397180 2% /run
/dev/sda1 48614564 10489612 35623064 23% /
tmpfs 2008016 12 2008004 1% /dev/shm
tmpfs 5120 8 5112 1% /run/lock
tmpfs 1024 0 1024 0% /run/credentials/systemd-journald.service
/dev/loop1 65408 65408 0 100% /snap/core20/2717
/dev/loop0 49280 49280 0 100% /snap/snapd/25935
/dev/loop2 15104 15104 0 100% /snap/cheat/4279
tmpfs 2008016 16 2008000 1% /tmp
tmpfs 401600 184 401416 1% /run/user/1000
frandyta@cis106:~$
Open
challenge_lab4.sh
~/cis106/labs/lab4
#!/bin/bash
echo
figlet "system"
echo -n "Date:"; date
echo -n "Hostname:"; hostname
echo -n "Uptime:"; uptime
echo -n "User:"; uname
figlet "MEMORY"
free
figlet "DISK"
df
echo

Script